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Assessment of bacteriological quality and presence of antibiotic resistant bacteria in vended sachet-packaged drinking water: potential threat of transmission of enteric pathogens and implications for public health

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Abstract Water in sachets is readily available and affordable, but there are concerns about its purity. With a view to ascertaining the bacteriological quality of sachet water in Coimbatore, five different brands of sachet water were procured from different localities in Coimbatore, and subjected to bacteriological analysis. The mean viable counts of heterotrophic bacteria in sachet water varied between 1.858 x 10^6 CFUs/mL to 2.333 x 10^4 CFUs/mL. The total coliform counts varied from a maximum of 1.366 CFUs/mL to complete absence of the indicator organism in a particular sample. Fecal Streptococci (FS) counts varied from 0 - 0.833 CFUs/mL. Fecal coliforms (FC) were not found in one sample, but were detected in all the other samples, with maximum counts of 2.8 CFUs/mL. The fecal coliforms were positively correlated with counts of fecal Streptococci. All the samples do not comply with ICMR and WHO standards. The levels of pathogens such as *Salmonella sp.*, *Vibrio sp.* and *Shigella sp.* detected in this study are of critical concern, since counts as high as 1.467 CFUs/mL, 0.45 CFUs/mL, and 1.733 x 10^5 CFUs/mL, were recorded for the three pathogens respectively. Significant correlation was observed among the counts of *Vibrio sp.* and *Salmonella sp*. Bacteria from the sachet water samples were resistant to three out of six antibiotics tested, namely, Ampicillin (10 mcg), Sulphatriad (300 mcg), and Penicillin G (1 unit). The multiple drug resistant (MDR) bacterial isolates characterized in this study are two strains of *Corynebacterium sp.* and twelve strains of *Bacillus sp*. The health implications to the general public on consumption of this water may be serious and calls for the intervention of the appropriate control agencies.

Key Words *Fecal Coliforms, Fecal Streptococci, Fecal Indicator Bacteria (FIB), Multiple Drug Resistance (MDR)*

Introduction

Water is the elixir of life (Usharani *et al*., 2010), and the bloodstream of the biosphere (Falkenmark and Rockstrom, 2004). Accessibility and availability of fresh clean water is necessary for sustainable development and an essential element in health, food production and poverty reduction. Good quality water is odorless, colorless, tasteless, and free from fecal pollution (Ezeugwunne *et. al*., 2009). Water pollution is a major cause of illness (Craun, 1988). The intake of unwholesome water could have devastating effects on our health as unsafe drinking water is a key determinant of many microbial diseases with serious complications in immuno compromised individuals (Steiner *et al*., 1997). Water contamination has been implicated as the causative factor in diseases such as typhoid, cholera, bacillary dysentery, etc.

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According to the World Health Organization, there are an estimated 4 billion cases of diarrhea and 2.2 million deaths annually (WHO, 2002) due to consumption of contaminated water.

The introduction of sachet water into the market by suppliers was to provide an alternative source of good quality water for consumption. Sachet or packaged drinking water is any water that is in a sealed plastic bag, distributed or offered for sale, and is intended for human consumption (FDA, 1995). Sachet water was supposed to be safe, hygienic, affordable and an instant source of water for consumers (Ahima and Ofosu, 2014). The appeal of the sachet seems to mirror the mass consumer appeal of small unit sizes of commodities as seen elsewhere in the developing world (Hammond and Prahalad, 2004).

Water in sachets is readily available and affordable, but there is concerns aboutits purity. The integrity of the hygienic environment and conditions where the majority of the water in sachets are produced has been questioned (CAMON, 2007). Potential health problems may occur due to the microbial content of the sachet water since water is one of the vehicles for the transmission of pathogenic organisms (Brock, 1991; Prescott *et al*., 2005). The bacteriological quality of drinking water is of paramount importance and monitoring must be given highest priority, because studies have attributed several disease outbreaks to untreated or poorly treated water containing bacterial pathogens.

Traditional assessments of microbiological water quality have focused on the presence of fecal indicator bacteria (FIB). FIB is abundant in human and other animal waste products where pathogenic organisms are found (Noble *et al*., 2003). Fecal Indicator Bacteria (FIB) indicates that a contamination pathway exists between a source of bacteria (surface water, septic system, animal waste, etc.) and the water supply. Disease-causing bacteria may use this pathway to enter the water supply. Some of the important pathogens include, *E. coli* O157:H7, *Shigella sp*., *Vibrio cholera* and *Salmonella sp.*

Total and fecal coliform bacteria have been used extensively for many years as indicators for determining the sanitary quality of water (Noble *et al*., 2003), and have been included in water quality standards in different parts of the world (ICMR, 1975; European Union 75 ⁄ 440 ⁄EEC 1975; USEPA, 1986 Water Quality Criteria; WHO, 1993).The maximum permissible value of total coliforms in drinking water is 1 per 100 ml (see ICMR, 1975), and 10 per 100 ml (WHO, 1993).

Fecal Coliforms (FC) are defined as those coliforms which respond at an elevated temperature of 44.5°C and are specific to the intestinal tracts of warm-blooded animals, including humans. The fecal coliform group comprises bacteria such as *Escherichia coli* or *Klebsiella pneumoniae*. Fecal *Streptococci* (FS) commonly inhabit the intestinal tract of humans and warm-blooded animals, and are also used to detect fecal contamination in water. High amounts of fecal streptococci indicate that the contamination is of animal origin. The group comprises *Streptococcus faecalis*, *S. bovis*, *S. equinus* and *S. avium*.

Several studies have documented the detection of high counts of heterotrophic bacteria, coliforms and pathogens in sachet drinking water which far exceeded the international standards set for potable water for human consumption. Assessment of the microbiological quality of sachet drinking water sold on the streets of Metropolitan

Kumasi, Ghana, revealed that 4·5% of the factory-bagged sachets contained total coliforms and 2-3% fecal coliforms, while counts of heterotrophic bacteria ranged from 2 to $6{\cdot}33 \times 10^5$ for factory-bagged sachet water (Obiri-Danso *et al.*, 2003). Forty-five percent (45%) of the brands of sachet water samples subjected to bacteriological examination in the Cape Coast municipality of Ghana contained coliform bacteria (Dodoo *et al.*, 2006). Onifade and Ilori (2008) reported the presence of *Alcaligenes faecalis*, *Bacillus subtilis*, *Enterobacter aerogenes*, *Escherichia coli*, *Micrococcus luteus*, *Staphylococcus aureus* and *Streptococcus faecalis* in sachet water samples purchased from different vendors in Akure, Ikare, Ondo and Owo in Ondo state, Nigeria. All brands of sachet water (100%) sampled by Oyedeji *et al.* (2010) from Ibadan and Ile-Ife cities in South Western Nigeria, had total coliforms, four brands (20%) had presence of *E. coli* while *Enterococcus faecalis* was recovered from two (10%) of the brands. Similarly, sachet water brands on sale in Bolgatanga municipality, Ghana, were found to be contaminated with coliform bacteria and fecal coliform bacteria (Oyelude and Ahenkorah, 2012). Bacteriological analysis of sachet water brands marketed in Maiduguri Metropolis, Borno State showed that 55% of the brands had fecal coliforms, 25% had *Pseudomonas aeruginosa*, 15% had *Salmonella sp*. while 5% of the brands had *E. coli* (Muazu*et al.*, 2012). Madukaet *al.* (2014) found prevalence of 42.43 % *E. coli*, 24.24 % *Streptococcus faecalis*, 12.12 % *Klebsiella pneumoniae*, and 21.21 % *Staphylococcus aureus*, in sachet water samples. Dada (2009) studied the bacterial quality of sachet water sold in Lagos, Nigeria. He observed that microbial quality deteriorated considerably as products moved farther down the distribution chain.

33.3% of the sachet water sampled by Venkatesan and Victor (2014) in Chennai, failed to meet the WHO drinking water standard making them unsuitable for human consumption The bacteria that were isolated included *Klebsiella pneumonia*, *Enterobacter aerogenes*, *Citrobacter freundii*, *Pseudomonas sp.*, *Acinetobacter sp.*, *Staphylococcus aureus*, *Coagulase negative Staphylococci*, *Micrococcus sp.* and *Bacillus sp*.

The emergence of bacteria resistant to most of the commonly used antibiotics/drugs is of considerable medical significance (Khan and Malik, 2001), because of the public health implications. There are several reports on the incidence of bacterial resistance among bacterial isolates obtained from food materials (Grewal and Tiwari, 1990; Singh *et al*., 1995), milk and milk products (Khan and Malik, 2001), animals such as chicken carcasses (Tessi *et al*., 1997), and sewage and water (Malik and Ahmad, 1994). It is common that bacteria that develop resistance to one antibiotic may also have the ability to develop resistance to another antibiotic which is usually referred to as multiple-antibiotic resistance (Walter and Vennes, 1985).

Antibiotic resistance has been reported for bacteria in sachet drinking water: The antibiotic susceptibility profile of the bacterial isolates from vended sachet water in Ogbomoso, Nigeria, revealed that, 59.30% were sensitive to commercial antibiotics, while 40.70% were resistant (Oladipo *et al*., 2009). Bacteria isolated by Banu and Menakuru (2010) from sachet water were resistant to antibiotics like amoxiclav, methicillin, chloramphenicol and streptomycin.

Multidrug resistant (MDR) bacterial strains pose a significant threat to public health

as treatment of diseases caused by these microbes is prolonged. Informationon the prevalence of resistance to specific drugs in bacterial species, together with changes occurring over time, are necessary to understand the magnitude of the problem and to establish baselines for taking action (Caprioli *et al*., 2000).

In order to safeguard public health it is essential that the available packaged water is of the highest quality. There is a need for strict and routine monitoring of packaged drinking water with a view to raising their standards. An understanding of their microbiological quality and safety are therefore pertinent. Hence, this study was carried out to assess the microbiological quality and presence of antibiotic resistant bacteria in vended sachet-packaged drinking water in Coimbatore, Tamil Nadu, and its potential impact on public health.

Materials and methods

Study Area

The current study was undertaken by procuring drinking water sachets sold in five different localities of Coimbatore. The selected localities are crowded areas frequented by people from all walks of life. Sampling area A (Near Railway Station) and B (Near Government Hospital) are located in the heart of Coimbatore city. The sampling sites C (Gandhipuram) and D (Gandhipark) are busy bus terminals having heavy commuter traffic, while E (Kalveerampalayam) is a rapidly developing suburb situated near Bharathiar University.

Collection of samples

The sachet water samples of varied brands were bought from vendors at the sampling locations, labeled, and brought to the laboratory for microbiological analysis. Bacteria present in the samples were then isolated and enumerated using standard microbiological methods (Cappuccino and Sherman, 1992).

Media and chemicals

The selective media and chemicals used in the study were procured from Hi Media Laboratories, Mumbai.

Microbiological Analysis

The total heterotrophic bacterial counts (THBC) of the sachet water samples were enumerated on nutrient agar medium using the serial dilution method and spread plate technique. The method of Dubey and Maheswari (2004) was adopted for the inoculation of media. Endo Agar (Endo, 1904) was used to confirm the detection and enumeration of coliform bacteria. M-FC Agar Base, designed by Geldreich*et al.* (1965) is used for the detection and enumeration of fecal coliforms using the membrane filter technique. Bile Esculin Azide Agar, a modification of Bile Esculin Agar (Swan, 1954; Facklam& Moody, 1970) as per Isenberg (1970), uses esculin hydrolysis in the identification offecal Streptococci or Enterococci. HiCrome RajHans Medium, Modified a modification of the original formulation of Rambach (1990), was used for differentiation of *Salmonella* species. TCBS Agar was used for the selective isolation and cultivation of *Vibrio cholera* and other enteropathogenic *Vibrio sp.* Shigella Base was used for the isolation and cultivation of *Shigella* species.

Antibiotic sensitivity / resistance of bacteria in sachet water samples

The sensitivity of the bacterial isolates from sachet water samples to six different antibiotics was determined by the Kirby-Bauer

method (1966). For this study, commercially available hexadiscs impregnated with six different antibiotics were procured from Hi Media Laboratories, Mumbai. The six discs contained, Ampicillin 10 mcg (Amp 10), Chloramphenicol 25 mcg (C 25), Penicillin G 1 unit (P 1), Streptomycin 10mcg (S 10), Sulphatriad 300 mcg (S_3 300), and Tetracycline 25 mcg (TE 25), respectively.

Isolation and Pure Culture of Antibiotic Resistant Isolates

Resistant bacterial colonies growing near the antibiotic discs on the surface of an agar plate culture were picked up with a sterile inoculation loop and transferred to separate nutrient agar plates. The isolates were further sub-cultured to obtain pure cultures.

Biochemical characterization of resistant Bacterial isolates

The drug resistant isolates were characterized using standard methods from Bergey's Manual of Determinative Bacteriology (Holt *et al.*, 1994). The characterization tests included, Gram's staining, endospore staining, starch hydrolysis test, indole test, Methyl Red (MR) test, Voges-Proskauer (VP) test, citrate utilization test, Triple Sugar Iron (TSI) test, glucose fermentation test, lactose fermentation test, sucrose fermentation test, maltose fermentation test, mannitol fermentation test, and catalase test.

Result and Discussion

Recent findings have highlighted the need for further evaluation of the microbiological quality of vended sachet water and raised concerns about its public health safety. With a view to ascertaining the bacteriological quality of sachet water in Coimbatore, five different brands of sachet water were procured from different localities in Coimbatore, and subjected to analysis.

The mean viable counts of heterotrophic bacteria in sachet water sampled in this study varied between 1.858 x 10^6 CFUs/mL to 2.333 x 10^4 CFUs/mL (Table 1). All the samples do not comply with the WHO and USEPA, which stipulate that the Maximum Contaminant Level (MCL) should not be more than 100 colonies and, any water harboring above 100 colonies should not be considered as safe water. This implies that the purification processes were improper, and hence, the water is not fit for human consumption.

The total heterotrophic bacterial counts in this study were considerably higher than those reported by other authors. Assessment of the microbial quality of sachet waters used for drinking in a typical rural community in Nigeria showed that, heterotrophic bacterial counts ranged from 10^1 to 10^7 CFUs/mL (Anthony *et al.*, 2005). The mean microbial counts in sachet water from Chennai were 1.1 x 10^2 CFUs/mL to 1.6 X 10^6 CFUs/mL (Venkatesan and Victor, 2014). The nature and quantity of microbes in the water samples determines the health risk to the consumer after consumption of the water containing pathogens and coliforms (Chan *et al.*, 2007).

The total coliform counts in the current investigation varied from a maximum of 1.366 CFUs/mL to complete absence of the indicator organism in a particular sample. Fecal *Streptococci* (FS) counts varied from 0-0.833 CFUs/mL. Similarly, fecal coliforms (FC) were not found in one sample, but were detected in all the other samples, with maximum counts of 2.8 CFUs/mL (Table 1). The fecal coliforms were positively correlated with counts of fecal *Streptococci* (Table 2).

The level of coliforms exceeded the maximum permissible values of 1 per 100ml (ICMR, 1975) in drinking water and 10 per 100 ml (WHO, 1993).

Coliforms and fecal coliforms have also been detected in sachet water elsewhere.Coliform counts of 1.62 log CFU/mL were recorded insachet-packaged drinking water from Western Nigeria (Olaoye and Onilude, 2009). Analysis of sachet water brands in Bolgatanga municipality, Ghana, revealed that, three out of every four sachet water samples were contaminated by coliform bacteria, ranging from 12-168 CFUs/100 mL (Oyelude and Ahenkorah, 2012). In Chennai city, India, the mean total coliform counts in sachet water ranged from 1.6 to 12.6 CFUs/100 ml (Venkatesan and Victor, 2014). Fecal coliform bacteria ranging from 2-63 CFUs/100 mL were detected in more than 60% of the sachet water brands on sale in Bolgatanga, Ghana (Oyelude and Ahenkorah, 2012).

Most coliforms are present in large numbers among intestinal flora of humans and other warm-blooded animals, and are thus found in fecal wastes (APHA, 1992). Consequently, coliforms, detected in higher concentrations than pathogenic bacteria, are used as an index of the potential presence of entero-pathogens in water. Positive presence of coliforms in treated water which is usually coliform-free, may indicate treatment ineffectiveness (Chan *et al*., 2007). Treated water should not contain these organisms, because they are indicator microorganism for other pathogens that may be present in feces (USEPA, 2003). In the case of sachet water, the sources of contamination could be the main water source because it is reported that some unscrupulous producers just bag and seal tap water without any form of treatment (Nwosu and Ogueke, 2004).The presence of fecal coliform observed in packaged water has been reported to be due to poor hygienic practices of producers, failure to wash hands, ignorance about good hygienic practices as well as the presence of animals in the vicinity of the factory (Coroler *et al*., 1999).

When water supplies contain coliform bacteria at levels greater than one per 100mL, the water may also contain pathogens that cause acute intestinal illness. The levels of pathogens such as *Salmonella sp.*, *Vibrio sp.* and *Shigella sp.* detected in this study are of critical concern, since counts as high as 1.467 CFUs/mL, 0.45 CFUs/mL, and 1.733 x 10^5 CFUs/mL, were recorded for the three pathogens respectively (Table 1). Significant correlation was observed among the counts of *Vibrio sp.* and *Salmonella sp.* (Table 2). The use of such water for drinking poses a serious threat to the health of the consumers and calls for the intervention of the appropriate control agencies.

Seven serotypes of *Salmonella sp.* were isolated from sachet water in Lagos, Nigeria – *Salmonella typhi* (8.1%), *Salmonella typhimurium* (21.6%), *Salmonella choleraesuis* (13.5%), *Salmonella enteritidis* (24.3%), *Salmonella paratyphi* (21.6%) and *Salmonella arizonae* (10.8%). Thirteen resistance patterns were exhibited by the isolates, with Str. Amp. Tet. Chl. Amo. Gen, Str. Amp. Tet. Amo. Chl. Amo. Nal. Nit, Str. Tet and Str. Amp. Tet. Chl. Amo being the most notable resistance patterns observed. Emerging multidrug-resistant *Salmonella* serotypes with *stn* gene may pose a threat to public health (Akinyemi *et al.*, 2011).

The results of this study indicate that bacteria from the sachet water samples were resistant to three out of six antibiotics tested, namely, Ampicillin (10 mcg),

Sulphatriad (300 mcg), and Penicillin G (1 unit) (Table 3; Fig. 1 ; Fig. 2). The presence of antibiotic resistant organisms in drinking water is significant, not only because they prolong the treatment of water borne diseases, but also because they pose the danger of promoting multiple antibiotic resistant organisms in humans.

Sachet water samples from Eastern Nigeria harbored 80.4% bacterial isolates resistant to ampicillin, 77.2% resistant to tetracycline and 23.7% resistant to ofloxacin (Nwachukwu and Emiruem, 2007). 40.7% of water borne bacterial pathogens isolated from sachet water vended in Ogbomoso, Nigeria, were resistant to antibiotics, while 59.3% were sensitive (Oladipo *et al.*, 2009).

The natural cause of antibiotic resistance is mutation or R-plasmid exchange between bacteria of the same species (Bell *et al*., 1980). Bacteria become resistant to antimicrobial agents by a number of mechanisms which are; production of enzymes which inactivate or modify antibiotics, changes in the bacterial cell membrane, preventing the uptake of antibiotics and development of metabolic pathways which enable the site of an antimicrobial action to be by-passed (Abbar and Kaddar, 1991). The common use of antibiotics in poultry or agricultural practices to prevent diseases can contaminate surface and underground water (El-Zanfaly*et al.*, 1988), from where they can be transferred to humans in drinking water, thereby adding to the resistance problem. Many antibiotics are persistent in the environment (Zuccato *et al*., 2000), and some of them have been isolated from wastewater and groundwater (Thurman and Hostetler, 2000). This could enhance the resistance of bacteria to antibiotics or drugs, and also spread bacterial resistance (Lateef *et al.*, 2005). Inappropriate use of antibiotics in preventing or treating human infections is believed to be the common cause of bacterial resistance. Antibiotic resistant bacteria are a cause for concern because of possible colonization of the gastrointestinal tract and conjugal transfer of antibiotic resistance to the normal flora leading to more multiple antibiotic resistant organisms (McKeon *et al.*, 1995). When these antibiotic resistant bacteria are introduced into water through fecal contamination, the emerging diseases will hardly respond to treatment.

The multiple drug resistant (MDR) bacterial isolates characterized in this study are two strains of *Corynebacterium sp.* and twelve strains of *Bacillus sp.* (Tables 4 & 5). Oladipo *et. al.* (2009) also isolated and identified *Bacillus subtilis*, *Bacillus alvei*, and *Bacillus cereus* from sachet drinking water. The presence of *Bacillus sp*. in the sachet water could be a result of contamination from poor staff handling during processing of the water samples (Okonko *et. al*., 2008). *Bacillus sp*. produces enterotoxin which could be deadly when ingested into the body.

The poor microbiological quality of vended sachet water may be due to several factors, notable among these are; improper processing and purification procedures, unhygienic handling after production, the small size of the pathogens which enable them to escape filtration and the resistance of these pathogens to physical water treatment agents and disinfectants (Cotte *et al.*, 1999). Packaged water products may be stored under direct sunlight. Coupled with the possibility of back-seepage into bags that are not well-sealed, the re-growth potential of microorganisms becomes significantly

increased owing to the high ambient temperatures (Hunter *et al*., 1990). Small scale producers of sachet water may depend on water from doubtful sources for the water they use in packing their products, some of them under very poor environmental conditions (Adekunle *et al*., 2004). Apart from environmental contaminants, contamination from improper vendor handling also poses threats to the health of the consumers who drink without proper cleaning of the sachets. Sachets are notorious for constituting a major proportion of the plastic waste generated, as consumers typically litter the plastic sleeves in streets and gutters.

The health implications to the general public on consumption of this water may be serious. The case is worsened by the isolation of fecal bacteria and pathogens, many of them exhibiting resistance to multiple antibiotics. Water borne diseases could be contracted and spread through drinking of such contaminated water.

There is need for the public to be properly informed of the presence of packaged drinking water of doubtful quality on the markets. The local regulatory authorities owe a duty to the public, to educate the vendors on hygiene, and to vend only government approved pure water sachets. The government should intensify efforts in the routine monitoring of activities in the packaged drinking water industry with a view to supply safe and wholesome water to the public. Assessment of water quality at different stages of production and post-production is therefore suggested in order to ensure the quality and safety. Regular monitoring and inspection will be required to enforce the existing regulations and if need be, promulgate new ones to ensure that the health of the populace is guaranteed.

Conclusion

This study shows that some sachet water on our markets could serve as possible routes of transmission of water-borne pathogens. Epidemiological and risk assessment approaches will undoubtedly boost our understanding of the occurrence, survival and transport of these organisms. Regulatory activities that promote core hygiene values and a proper handling culture could produce the desired improvements rather than a focus on end-product monitoring, which does not always give a complete picture in terms of microbiological risk assessment. There is a need for regulatory and health agencies to maintain a balanced position, that concurrently improves social welfare and access to safe drinking water.

Acknowledgement

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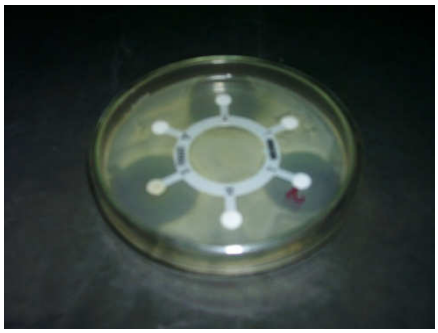

Fig. 1: Antibiogram of MDR Isolate

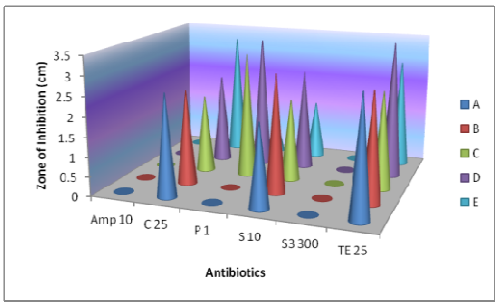

Fig. 2: Antibiotic Sensitivity of Bacteria in Sachet Water Samples

Table 1: Microbial contamination in sachet drinking water

Type of Microbe	Sample A	Sample B	Sample C	Sample D	Sample E
MVC of bacteria (THBC)	2.333×10^4	1.858×10^6	7.85×10^4	1.25×10^5	4.17×10^4
Total Coliforms	0.383	0	1.3	0.8	1.366
Fecal Coliforms	1.733	0	2.8	0.25	1.783
Fecal*Streptococci*	0.033	0	0.283	0.283	0.833
Salmonella sp.	4.983	0	1.467	1.017	0.383
Vibrio sp.	0.45	0	0.183	0.167	0.217
Shigella sp.	1.25	0	4.516	1.733×10^5	1.6×10^5

Table 2: Correlation matrix of microbiological parameters in sachet drinking water

	MHC	TC	FC	*Salmonella sp.*	*Vibrio sp.*	*Shigella sp.*
Total Coliforms	- 0.726					
Fecal Coliforms	- 0.647	0.686				
Salmonella sp.	-0.464	-0.190	0.362			
Vibrio sp.	-0.730	0.151	0.511	**0.908**		
Shigella sp.	- 0.386	0.468	- 0.255	- 0.392	- 0.070	
FS	-0.482	**0.819**	0.336	-0.399	0.015	0.712

Table 3: Antibiotic sensitivity of bacteria in sachet water samples

	Sample A	Sample B	Sample C	Sample D	Sample E
	Zone of Inhibition (cm)				
Amp 10	0	0	0	0	0
C 25	2.6	2.4	2.0	2.3	3.2
P 1	0	0	3.2	3.4	1.1
S 10	2.1	3.0	2.1	2.6	1.5
S_3 300	0	0	0	0	0
TE 25	3.0	2.8	2.5	3.5	2.8

Table 4: Biochemical characterization of MDR Isolates 1 to 7

MDR Isolate	Gram Staining	Cell morphology	Motility	Spore staining	IMViC				TSI				Starch Hydrolysis	Carbohydrates					Catalase	Organism
					Indole	MR	VP	Citrate	Butt	Slant	H2S production	Gas production		Glucose	Lactose	Maltose	Mannitol	Sucrose		
1	+	Rod in chain	-	-	-	-	-	+	K	K	-	-	-	-	-	-	-	-	+	***Cornybacterium sp.***
2	+	Rod in chain	-	+	-	+	-	+	A	A	-	-	+	+	+	+	+	+	-	***Bacillus sp.***
3	+	Rod in chain	-	+	-	+	-	-	A	A	-	-	+	+	+	+	+	+	-	***Bacillus sp.***

4	+	Rod in chain	-	+	-	+	-	-	A	A	-	-	+	+	+	+	+	+	-	***Bacillus sp.***
5	+	Rod, single	-	+	-	+	-	+	A	A	-	-	+	+	+	+	+	+	-	***Bacillus sp.***
6	+	Rod in chain	-	+	-	+	-	-	A	A	-	-	+	+	+	+	+	+	+	***Bacillus sp.***
7	+	Rod in chain	-	+	-	+	-	+	A	A	-	-	+	+	+	+	+	+	+	***Bacillus sp.***

Table 5: Biochemical Characterization of MDR Isolates 8 to 14

MDR Isolate	Gram Staining	Cell morphology	Motility	Spore staining	IMViC				TSI				Starch Hydrolysis	Carbohydrates					Catalase	Organism
					Indole	MR	VP	Citrate	Butt	Slant	H2S production	Gas production		Glucose	Lactose	Maltose	Mannitol	Sucrose		
8	+	Rod in chain	-	+	-	+	-	+	A	A	-	-	+	+	+	+	+	+	-	***Bacillus sp.***
9	+	Rod in chain	-	+	-	+	-	+	K	A	-	-	+	+	+	+	+	+	-	***Bacillus sp.***
10	+	Rod in chain	-	+	-	+	-	+	A	A	-	-	+	+	+	+	+	+	-	***Bacillus sp.***
11	+	Rod in chain	-	+	-	+	-	+	A	A	-	-	+	+	+	+	+	+	+	***Bacillus sp.***
12	+	Rod in chain	-	+	-	+	-	+	A	A	-	-	+	+	+	+	+	+	-	***Bacillus sp.***
13	+	Rod in chain	-	-	-	-	-	+	K	K	-	-	-	-	-	-	-	-	+	***Bacillus sp.***
14	+	Rod, short	-	+	-	+	-	+	A	A	-	-	+	+	+	+	+	+	+	***Cornybacterium sp.***

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